

WATSONVILLE, CA, USA

WMO: 745058

Lat: 36.936N

Lon: 121.789W

Elev: 160

StdP: 14.61

Time Zone: -8.00 (NAP)

Period: 99-19

WBAN: 23277

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) 34.1	(c) 36.4	(d) 22.0	(e) 16.6	(f) 52.5	(g) 25.9	(h) 20.0	(i) 50.1	(j) 18.4	(k) 56.2	(l) 15.7	(m) 56.8	(n) 2.5	(o) 330	(p) 0.455

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 9	(b) 20.5	(c) 86.2	(d) 62.5	(e) 81.1	(f) 62.0	(g) 77.0	(h) 61.2	(i) 65.4	(j) 79.8	(k) 64.1	(l) 77.1	(m) 62.8	(n) 74.4	(o) 6.4	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.9	80.3	66.6	59.3	75.9	66.2	58.3	73.1	65.3	30.2	80.1	29.2	77.0	28.4	74.4	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 15.0	(b) 12.4	(c) 10.9	(e) 30.1	(f) 96.8	(g) 2.6	(h) 4.3	(i) 28.3	(j) 99.9	(k) 26.8	(l) 102.4	(m) 25.3	(n) 104.8	(o) 23.4	(p) 107.9		
			DB													
			WB													
				28.2	68.9	2.5	1.4	26.4	70.0	24.9	70.8	23.5	71.6	21.7	72.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.6	51.3	52.3	54.2	55.6	58.4	61.2	62.6	63.3	63.8	61.2	55.5	50.9
	DBStd	6.43	4.89	4.97	4.70	4.67	3.91	3.58	3.09	3.48	4.93	5.18	5.10	4.61
	HDD50	148	45	29	13	5	0	0	0	0	0	0	11	45
	HDD65	2855	424	356	337	287	212	124	91	75	81	143	287	438
	CDD50	2906	86	94	142	173	260	337	389	414	413	348	177	73
	CDD65	137	0	0	1	5	6	11	15	24	45	27	3	0
	CDH74	1652	10	15	40	92	112	159	123	181	461	390	68	1
(13)	CDH80	549	1	0	6	33	34	51	20	53	193	145	13	0
(14)	Wind	WSAvg	4.1	3.3	4.0	4.2	4.7	4.9	4.8	4.4	4.1	3.7	3.2	3.4
(15) Precipitation	PrecAvg	21.3	4.2	3.5	3.6	1.9	0.2	0.1	0.1	0.1	0.3	1.1	3.3	3.5
	PrecMax	48.3	10.9	10.4	11.0	6.7	1.0	1.3	1.3	1.1	1.5	3.8	8.0	10.5
	PrecMin	10.7	0.2	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0
	PrecStd	8.1	2.9	3.0	2.4	2.0	0.2	0.2	0.3	0.2	0.4	1.1	2.4	2.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.0	76.2	79.4	84.0	84.2	86.7	82.2	86.6	93.4	91.1	81.5	71.6
		MCWB	54.5	58.8	62.1	62.9	63.0	63.8	65.3	65.3	64.7	62.1	60.3	53.7
	2%	DB	69.5	71.1	72.5	74.6	75.7	78.3	77.4	78.6	85.7	83.4	74.4	66.0
		MCWB	54.7	56.5	58.4	58.8	59.7	62.5	63.6	63.9	61.9	60.3	58.2	53.7
	5%	DB	64.2	65.4	67.5	69.2	70.8	73.2	73.3	74.1	78.7	77.4	69.6	62.7
		MCWB	53.8	54.5	56.2	56.3	58.8	60.8	62.1	62.6	62.4	59.0	55.9	53.3
	10%	DB	60.4	61.3	63.5	64.6	66.9	70.1	70.3	71.5	73.1	72.0	64.6	59.6
		MCWB	52.8	53.4	54.7	55.3	57.4	59.7	61.0	61.7	61.3	58.1	54.8	52.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.3	61.2	63.4	64.1	64.9	65.9	66.4	66.4	67.2	65.4	63.3	60.8
		MCDB	66.0	70.4	77.0	81.9	79.7	81.6	80.1	81.5	82.4	80.5	74.1	64.3
	2%	WB	57.1	58.7	60.1	60.5	61.3	63.4	64.1	64.5	64.9	63.1	60.6	58.2
		MCDB	62.7	66.4	69.3	71.7	73.2	76.3	76.0	76.7	79.4	76.1	68.0	60.9
	5%	WB	55.5	56.7	57.9	58.2	59.4	61.5	62.5	63.1	63.4	61.5	58.9	56.2
		MCDB	60.8	63.1	64.6	66.3	68.8	72.4	72.6	73.7	76.1	72.6	65.7	59.2
	10%	WB	54.0	54.7	56.0	56.3	57.9	60.0	61.2	61.8	61.8	59.8	57.1	54.2
		MCDB	58.8	59.9	61.7	63.2	66.1	69.1	69.8	70.8	72.1	69.0	63.2	57.7
(35) Mean Daily Temperature Range	5% DB	MDBR	19.1	18.1	18.7	19.1	18.6	18.7	16.8	17.0	20.5	22.1	21.0	18.5
		MCDBR	26.2	26.6	26.7	28.0	26.9	26.4	23.0	23.3	31.1	32.7	28.6	23.2
		MCWBR	14.7	14.8	14.9	14.9	13.8	12.7	10.8	10.8	13.1	14.1	14.1	13.3
	5% WB	MCDBR	19.4	20.6	22.0	23.2	23.1	23.1	21.2	21.7	25.9	25.5	21.2	14.1
		MCWBR	12.2	12.4	13.2	13.5	12.6	11.5	10.0	10.1	11.6	12.0	11.9	9.7
(40) Clear-Sky Solar Irradiance	taub		0.319	0.325	0.326	0.353	0.364	0.355	0.358	0.342	0.350	0.323	0.311	0.315
	taud		2.548	2.538	2.530	2.431	2.416	2.452	2.461	2.496	2.465	2.576	2.576	2.574
	Ebn at Noon		278	290	299	294	291	292	291	294	286	287	279	272
	Edh at Noon		24	28	31	36	37	36	35	33	32	26	23	23
(44) All-Sky Solar Radiation	RadAvg		809	1094	1543	1969	2279	2487	2345	2114	1803	1342	941	714
	RadStd		95	122	130	165	135	100	70	80	67	79	52	93

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.65	N/A	N/A	N/A	N/A	+1.26	N/A	-351	+331	N/A
(47)	Regional (44 neighbors)	+0.67	N/A	-2.20	N/A	N/A	N/A	N/A	-194	+245	+56

Nomenclature: See separate page